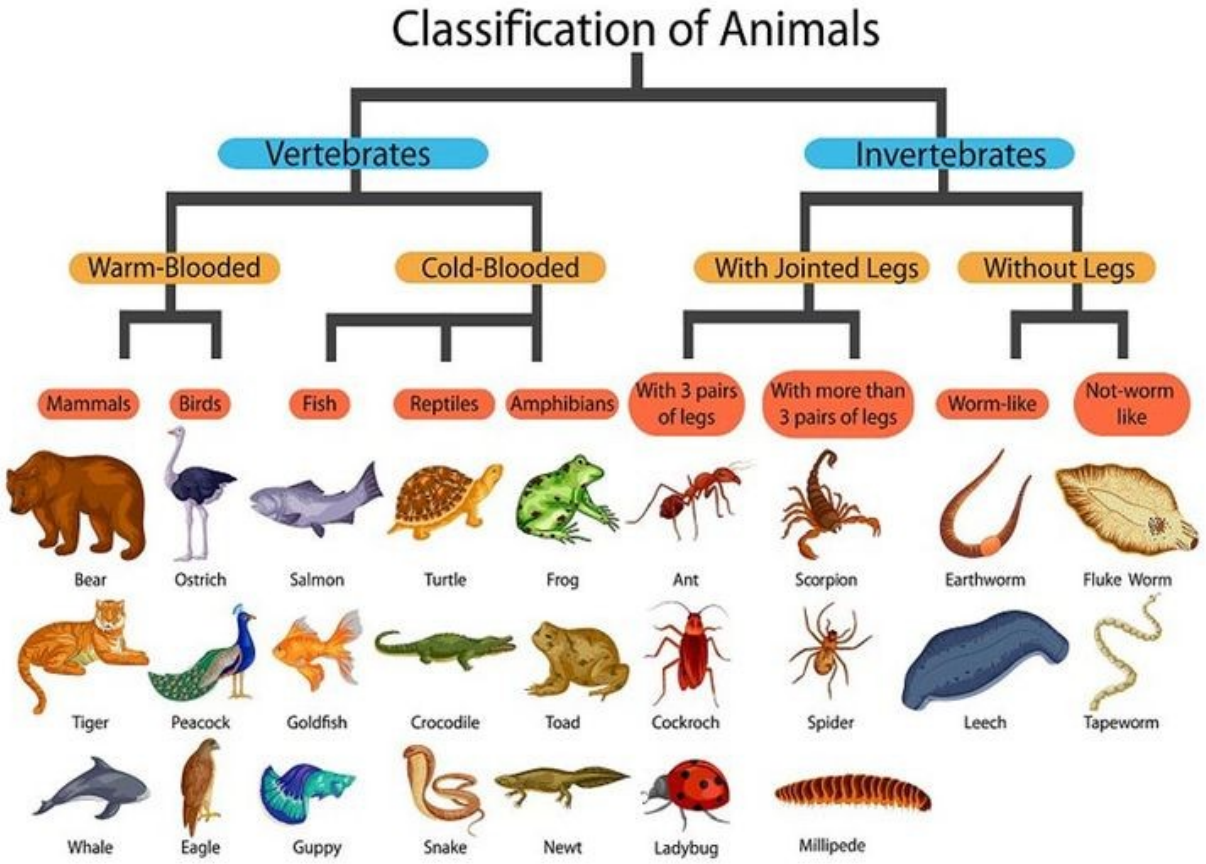
An Innovation Ecosystem to encourage Cybersecurity by Design?

Reid Derby, CyNam
12 Nov 2024

If you could be an animal, what type would you be?



Eurasian Beaver

Castor fiber

BEAVER TRUST

Although short-sighted it is believed beavers can see colour. They also have a third, clear eyelid, (nictating membrane) which allows them to see underwater.

An excellent sense of hearing both on land and underwater allows them to identify predators and family. The inner ear is full of dense fluffy hairs which trap air and prevent water from getting in.

Beavers have 2 layers of fur. A short dense underfur which acts as an insulator and an outer layer of longer guard hairs which repel water.

Vital for finding food, identifying family members and detecting danger, a beaver's sense of smell is it's most important and one of its strongest senses.

With a lung capacity roughly 3 times that of an adult human beavers can remain underwater for up to 15 minutes at a time.

104

Beaver fossils and other archaeological remains have been discovered at 104 different sites across Britain.

Up to
1.5 feet (0.5m)

Weigh up to:
30kg

Beavers teeth are reinforced with iron which gives them their distinctive orange colour and allows them to gnaw through wood.

Large, flat and scaly tail mainly used as a rudder when swimming. Also used to slap the water's surface to deter predators and notify other beavers of danger.

28 European countries have reintroduced beavers since the 1920's. The European population is estimated at 1.5 million individuals but Britain only has an estimated 2,000.

Up to
4.3 feet (1.3m)

Hand-like forepaws allow beavers to easily grasp food and materials. But unlike us it is the outer digit of the hand which is opposable (imagine your thumb and pinky swapped places).

Large webbed hind feet (up to 7 inches) which are mainly used to propel beavers through water up to speeds of 5mph.

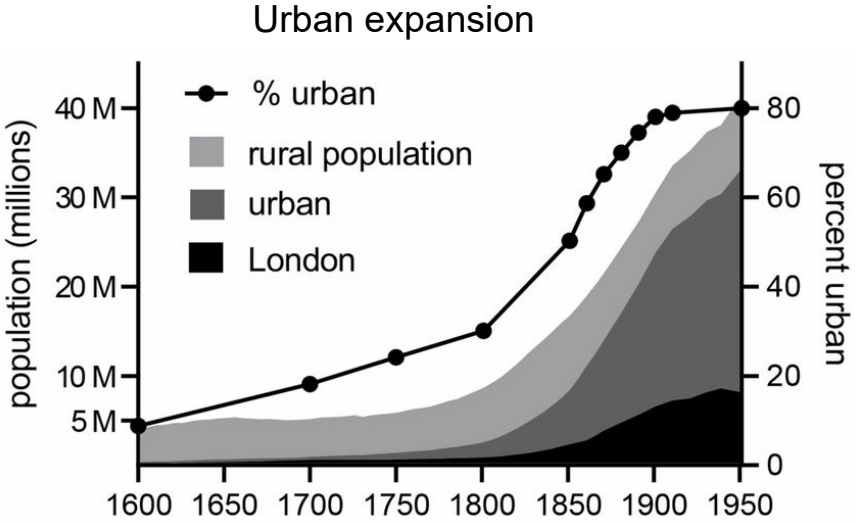
2nd
Largest rodent in the world

Out of 4,660 known species of rodent, the beaver is second only to the Capybara from South America in size.

What caused the decline of beavers in the UK?

Deforestation

Monoculture Farming



Beaver was hunted for meat and fu

And Castoreum – secreted from beaver castor sacs

By H. Zell - Own work, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=19881156

Depiction of a beaver hunt in the 13th Century Salisbury Bestiary, British Library

The Return of the European Beaver

Beavers have existed in Europe for more than 30 million years and populated almost the entire continent. But from the 12th century onwards, these large rodents experienced a sharp population decline due to the deterioration of their habitats and the extensive hunting of fur, meat* and castoreum. At the beginning of the 20th century, only about 1200 animals had survived in 8 isolated regions of Eurasia. Today, more than 1 million animals are estimated in Europe, and populations are increasing again.**

* During the Council of Constance (1414-1418) it was decided that beavers had enough resemblance to fish to be eaten during Lent. This resulted in a number of recipes with "beaver's tail", which was prepared like fish and eaten during fasting or on Fridays.

** Castoreum is a strong-smelling secretion from the beaver's castor sacks used as a medicine and basic aroma in perfumes. It is also believed to act as a potency enhancer and aphrodisiac.

Eurasian beavers are the largest rodents native to Eurasia. They weigh 11-30 kg and measure 105-150 cm from head to tail.
As a "water body designer", the beaver changes the landscape, creates habitats for numerous other species and improves water quality. In beaver regions there is usually a higher biodiversity of insects, amphibians, birds and fish. Because of these design activities, it is considered a "key species" of semiaquatic ecosystems.

In an effort to supplement the almost extinct beaver population, 7 North American beavers were released in Finland in 1937. By 1973 it turned out that North American beavers belonged to another subspecies (Castor canadensis) than Eurasian beavers (Castor fiber) with 40 instead of 48 chromosomes. Due to its slightly higher reproduction rate, the North American beaver is slowly displacing its Eurasian competitor. It is since discussed to eradicate the alien species.

290
animals survived along the Neman and Dnepr rivers (Belarus and Ukraine)

70
animals survived along the Don river (Russia)

100
animals survived in the Telemark region (South Norway)

200
animals survived along the Elbe (Central Germany)

30
animals survived along the Rhone (South France)

Another
500
animals survived in three separate populations in Siberia and China

- Historic distribution of Beavers (12th century)
- Relict populations (beginning of 20th century)
- Today's spread of the Eurasian beaver (Castor fiber)
- Today's spread of the North American beaver (Castor canadensis)

Sources:

Basemap: Natural Earth I Hillshade by Esri, Airbus DS, USGS, NGA, NASA, CGIAR, N Robinson, NCEAS, NLS, OS, NPA, Geodatastyrelsen, Rijkswaterstaat, GSA, Geoland, FEMA, Intermap and the GIS user community

Thematic data:

D. Halley, Norwegian Institute for Nature Research, Trondheim, personal communication

D. Halley, F. Rosell & A. Saveljev, 2018

G. Veron, 'Histoire biogéographique du Castor d'Europe, Castor fiber (Rodentia, Mammalia), 1992

Jonas Beinder, 2019

The effects of beaver reintroduction into an ecosystem

PHYSICAL CHANGES

Stream alteration
- Reduced current velocity
- Raised water table
- Increased water surface area & depth
- Increase in sediment trapping
- Reduction of turbidity downstream
- Improved regulation of stream acidity
- Diversification of river bank structure
- Increase in woody debris

Riparian zone alteration
- Felling of the trees
- Increase in open canopy area

EFFECTS ON BIODIVERSITY

- Creation of habitat mosaic
- Increase in pond, pond edge, and dead wood inhabiting species
- Provision of food source and cover for adult and juvenile fish
- Promotion of regeneration of target species (e.g. willow)
- Enhanced understory diversity in riparian zone

EFFECTS ON ECOSYSTEM FUNCTIONS

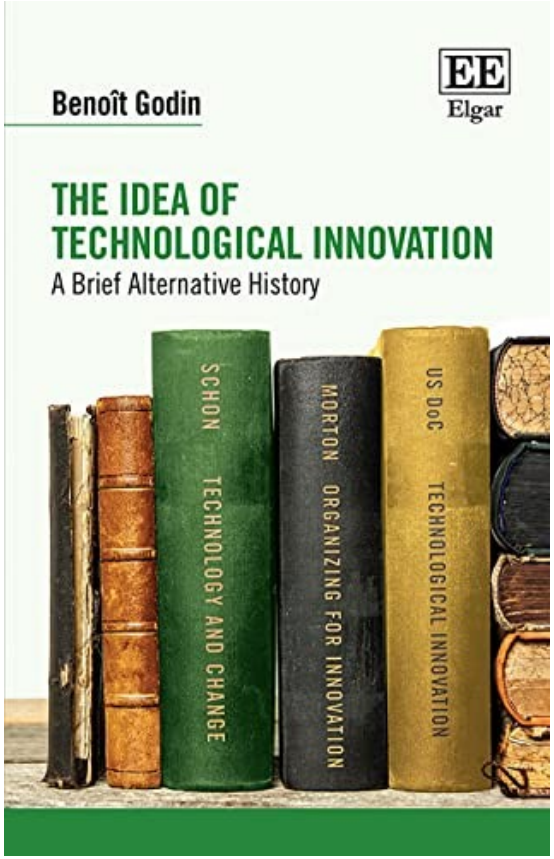
- Increased water storage
- Flood control
- Erosion control
- Increase in available N and P
- Increase in dissolved organic matter
- Enhanced nutrient cycling
- Mediation of water temperature
- Increased ecosystem resistance to disturbances

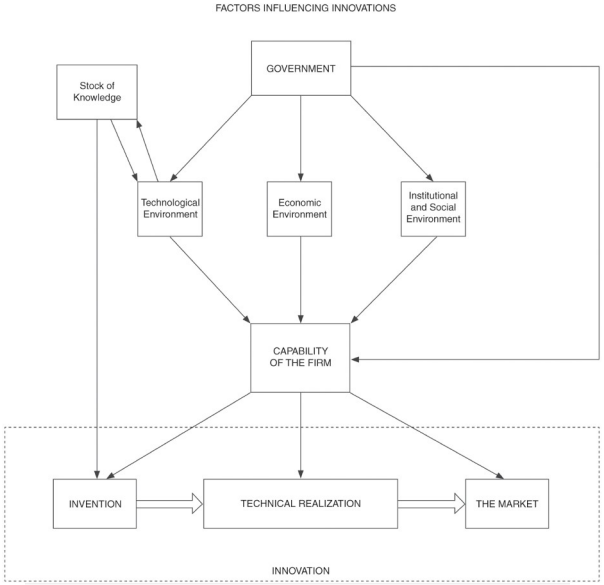
LETS TALK GEOGRAPHY

TYPES OF ECOSYSTEM

WHEN YOU TAKE A METAPHOR
LITERALLY
imgflip.com

What is an innovation (eco)system?





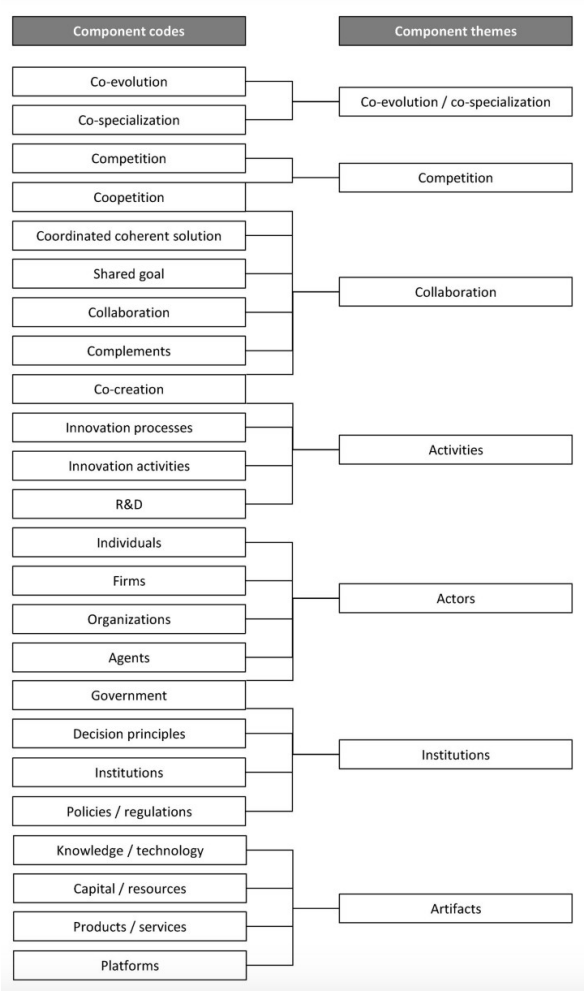
Harvard Business Review
www.hbrreprints.org

HBR SPOTLIGHT

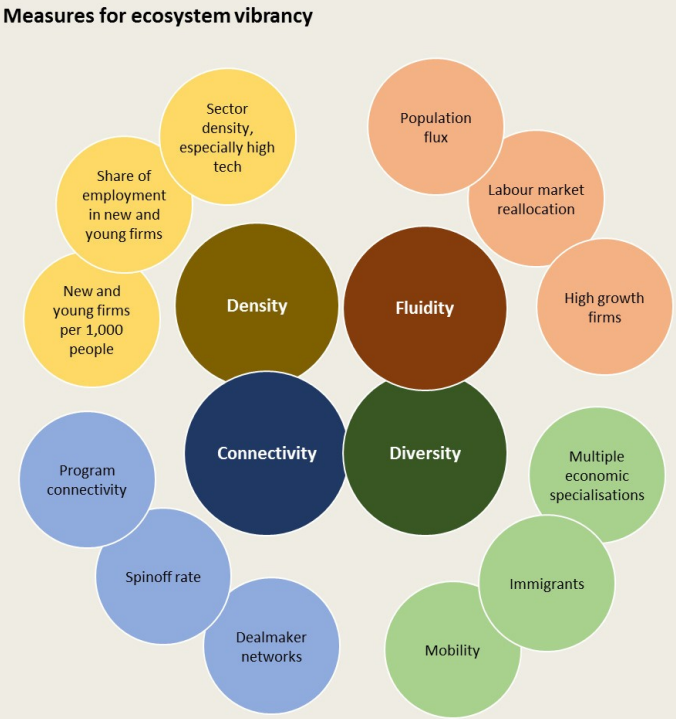
Successful innovation requires tracking your partners and potential adopters as closely as you track your own development process.

Match Your Innovation Strategy to Your Innovation Ecosystem

by Ron Adner

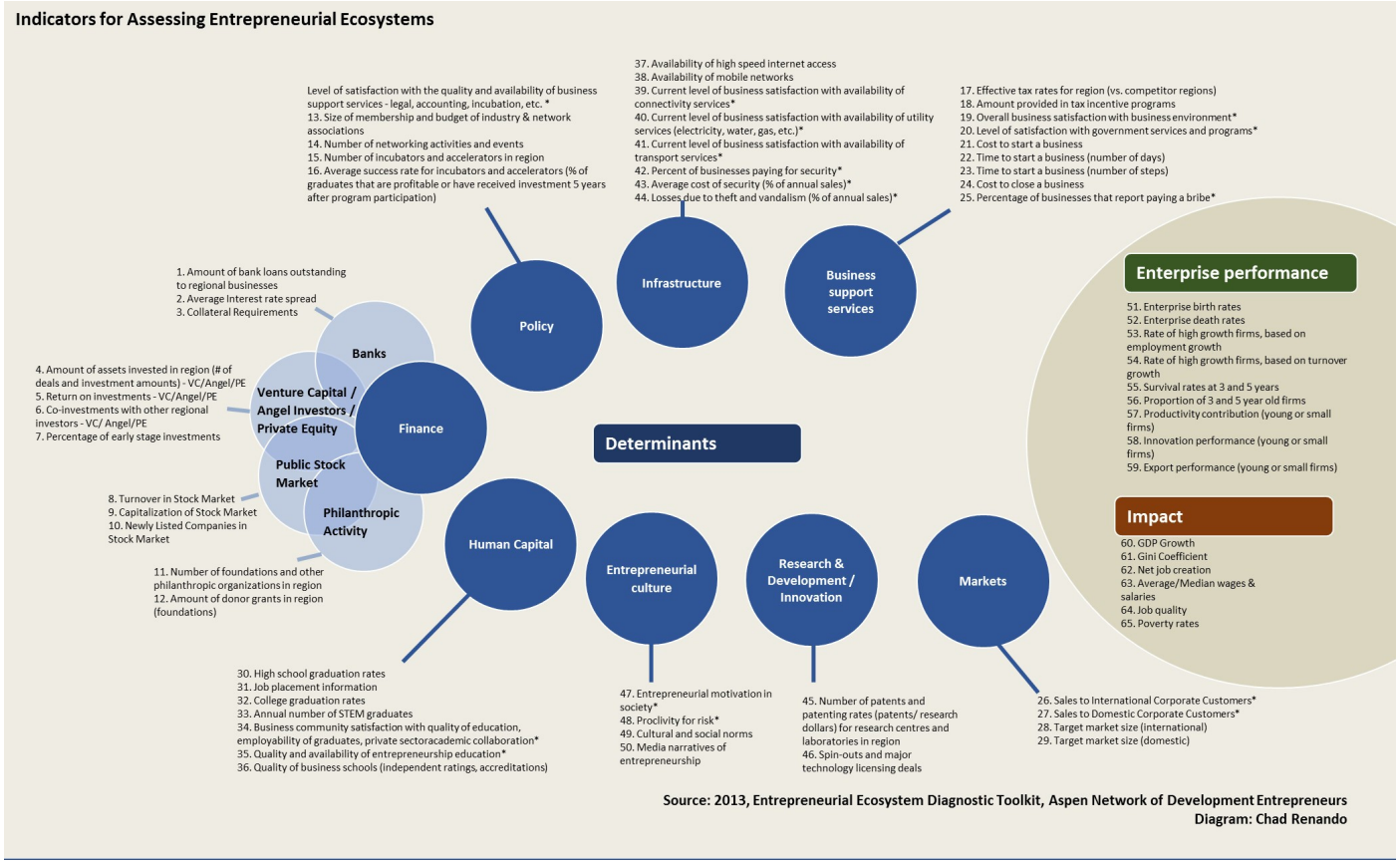


Model 1: Measuring Entrepreneurial Ecosystem Vibrancy, the Kauffman Foundation

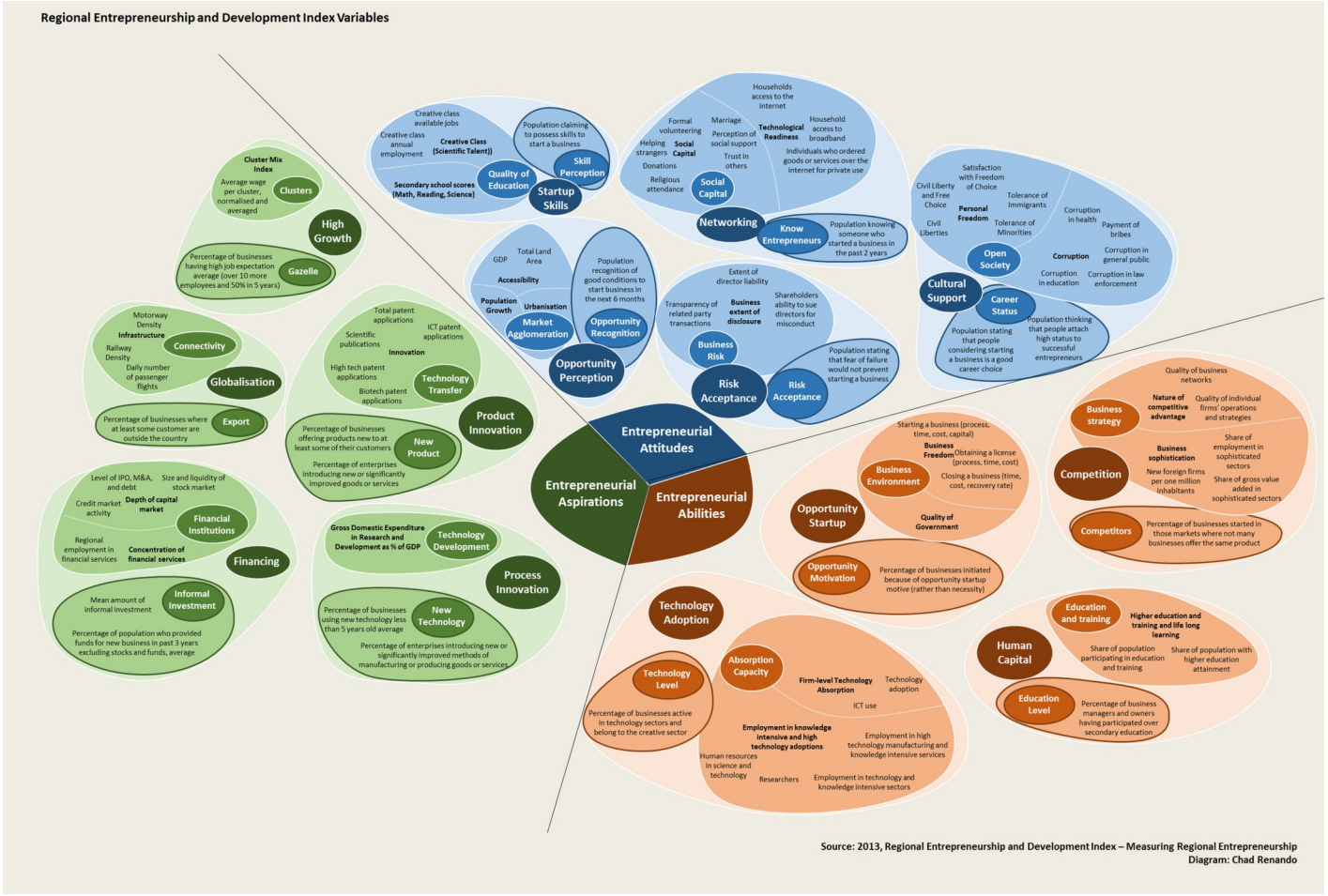


Source: 2015, Measuring an Entrepreneurial Ecosystem,
Ewing Marion Kauffman Foundation
Diagram: Chad Renando

Model 2: Entrepreneurial Ecosystem Diagnostic Toolkit, the Aspen Network of Development Entrepreneurs

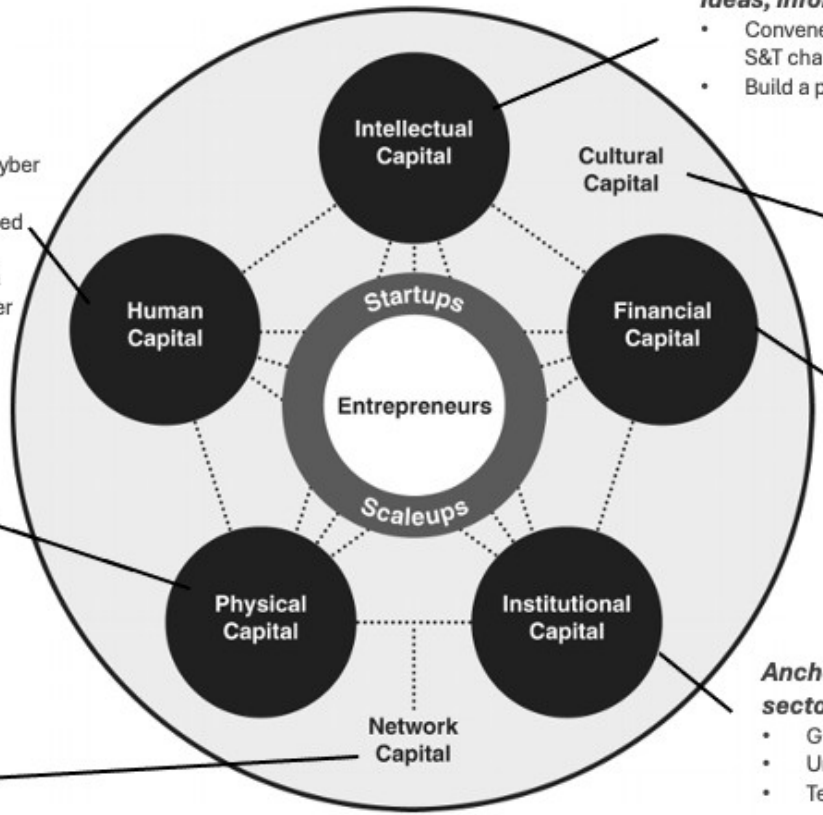


Model 3: The Regional Entrepreneurship and Development Index (REDI)



Create conditions & build capacity for a thriving entrepreneurial and innovation ecosystem: *The 7 Capitals*

Sits at the heart of a national ecosystem



Ideas, information, technologies, stories

- Convene research institutes around future national security and S&T challenge areas
- Build a platform for sharing stories of innovation, entrepreneurship

Talent, knowledge, skills, experience, diversity

- National Centre for Teacher CPD in Tech & Cyber (build on NCCE hubs e.g. Pate's Odyssey)
- Community engagement programme – focused on areas of high deprivation
- Enhance Entrepreneurial/Spin-out support & mentoring programme – e.g. Pivot Labs, Cyber Runway

Mindset, behaviours, history, inclusiveness, love of place

- Open, sharing & collaborative environment: #GiveFirst
- Interactive museum
- Chelt Festivals – Public Engagement on S&T

Infrastructure, facilities

- "Innovation Hotel"
- Event Space(s)
- Shared Lab Space
- Demo and PoC Environments

Investment, financing

- Pre seed fund – ready access to very early-stage finance to encourage Spin-outs
- Investment community engagement activities
- Angel Network

Connectedness, relationships

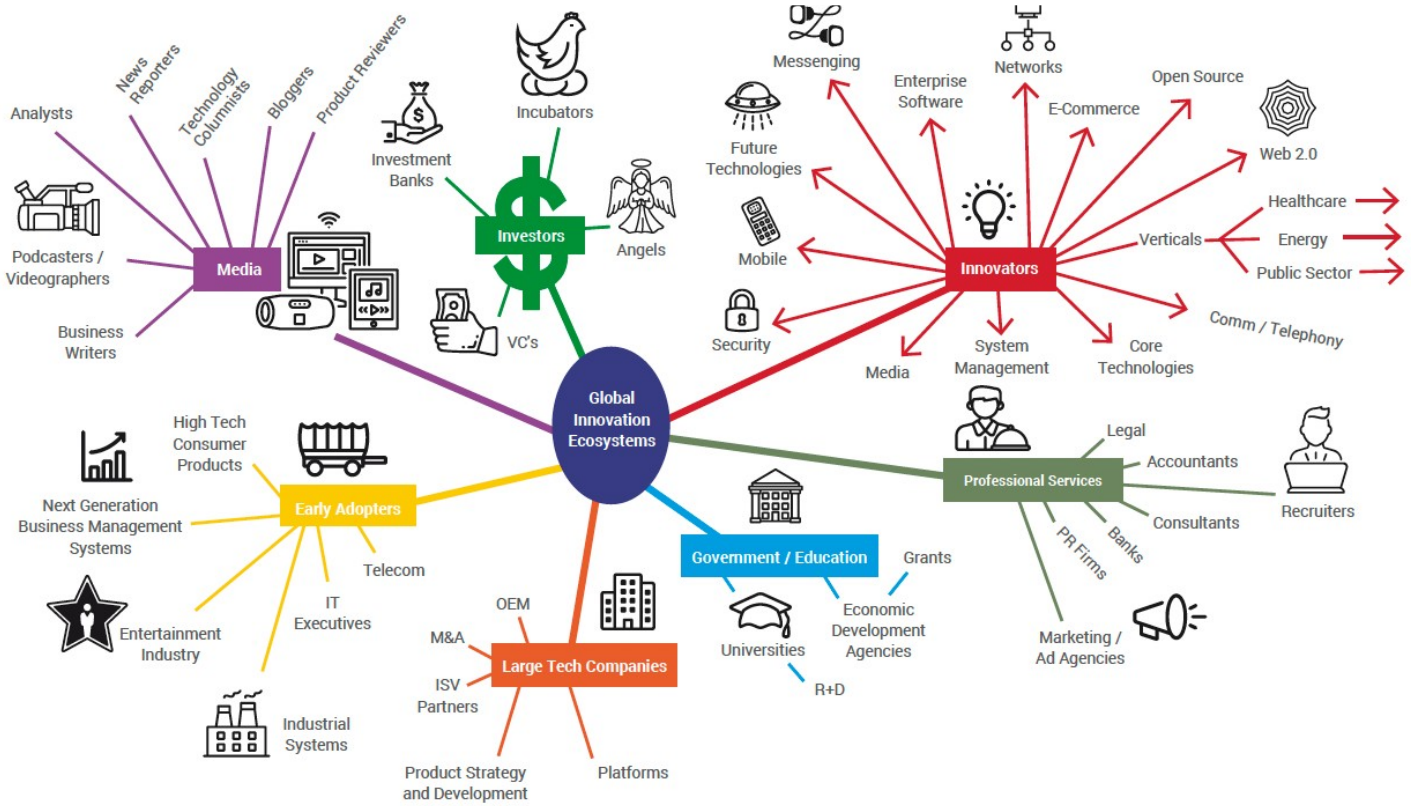
- "Ecosystem Immersion" offering
- Engagement events with:
  - Cyber Clusters
  - Em-tech Clusters
  - International Clusters

Anchor institutions, strong public sector, markets

- GCHQ/NCSC
- Universities
- Tech Primes

Brad Feld & Ian Hathaway: The Startup Community Way: Evolving an Entrepreneurial Ecosystem

Find, connect and mobilize different **people** from across the ecosystem …



Source: ‘The power of peers: The future of collaboration lies in global Innovation Ecosystems’ (Jennifer Vessels, May 2022)
https://www.globalfocusmagazine.com/the-power-of-peers-the-future-of-collaboration-lies-in-global-innovations-ecosystems/

We need Ecosystem Engineers to build the CHERI Ecosystem

(Source: ‘Beavers Work, If Permitted’ by Elliot McCandless, Sep 2019, https://beavertrust.org/beavers-work-if-permitted/)

How can we create a dynamic CHERI innovation network?

The Nature of Networks that Support Innovation

Tobias Stone @ Newsquare · Follow
10 min read · Oct 9, 2023

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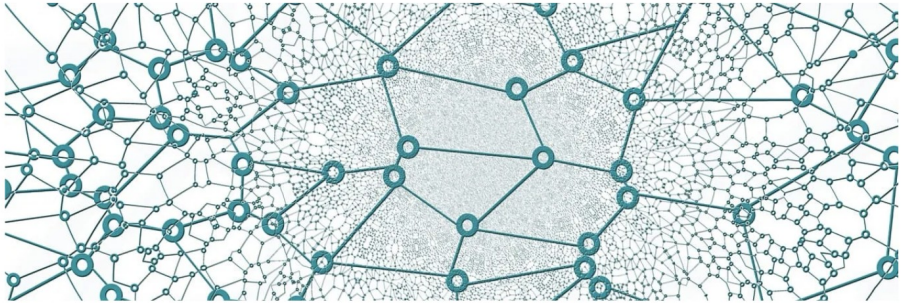


*"A **dynamic network** that has **strong ties** with a high level of closure at the core, set within a **weak network architecture** rich in **bridging ties**. It creates value by facilitating the flow of non-redundant information across structural holes, using the weak ties in its network. **Social Capital** is the main currency of the network, its value being embedded within its social network. The norms and rules of the network are policed using **link reciprocity**, which relies upon the social network being dynamic and rich in Social Capital."*

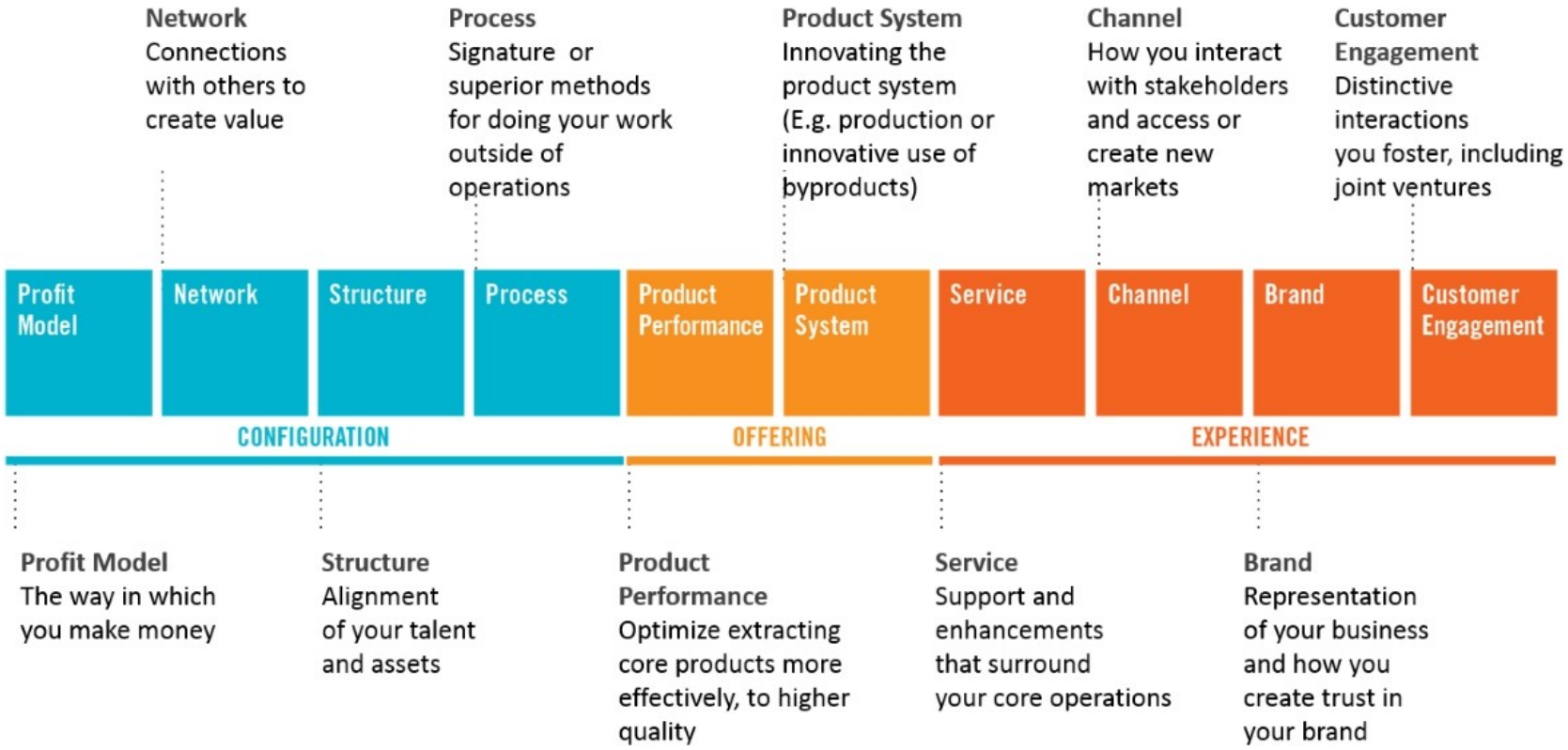
Toby Stone,'The Nature of Networks that Support Innovation' https://medium.com/@newsquare/the-nature-of-networks-that-support-innovation-e496784b61c3

We need CHERI Entrepreneurs

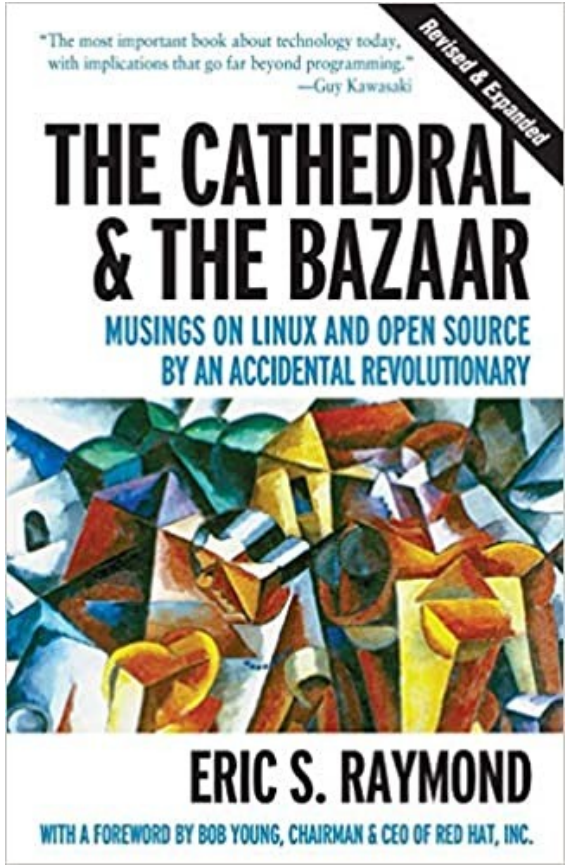
(Source: '10 Characteristics of Successful Entrepreneurs', Kelsey Miller, Jul 2020 https://online.hbs.edu/blog/post/characteristics-of-successful-entrepreneurs)

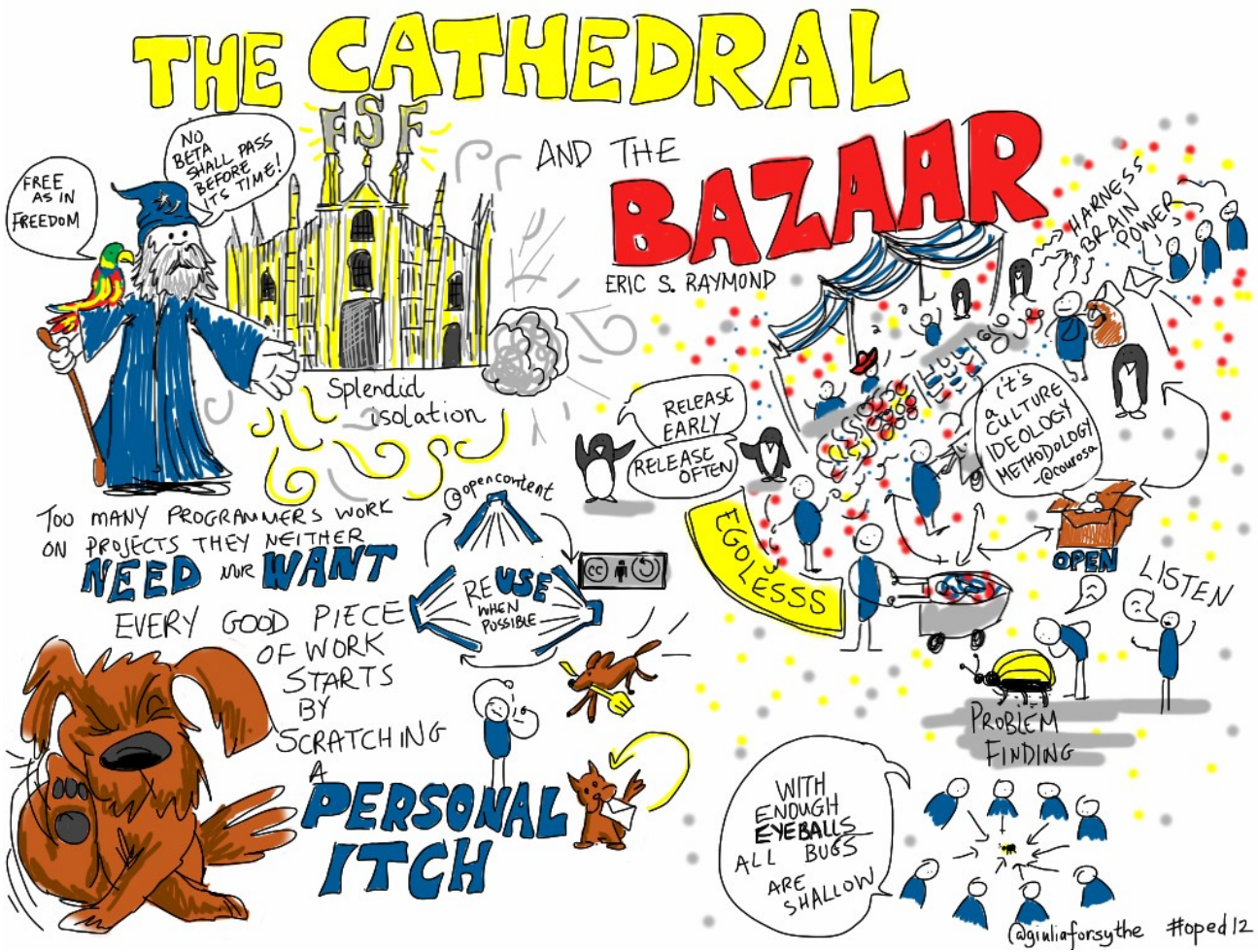
Harvard Business School Online

10 types of Innovation for CHERI?



(Source: Peter Fisk blog post on Innovation https://www.peterfisk.com/insights/key-topics/innovation/)


"The most important book about technology today, with implications that go far beyond programming."
—Guy Kawasaki
Revised & Expanded
THE CATHEDRAL & THE BAZAAR
MUSINGS ON LINUX AND OPEN SOURCE BY AN ACCIDENTAL REVOLUTIONARY
ERIC S. RAYMOND
WITH A FOREWORD BY BOB YOUNG, CHAIRMAN & CEO OF RED HAT, INC.


THE CATHEDRAL AND THE BAZAAR
ERIC S. RAYMOND
FSF
FREE AS IN FREEDOM
NO BETA SHALL PASS BEFORE IT'S TIME!
Splendid isolation
HARNESS BRAIN POWER
RELEASE EARLY
RELEASE OFTEN
it's a CULTURE IDEOLOGY METHODOLOGY -@cowrosa
EGOLESSS
OPEN
LISTEN
PROBLEM FINDING
TOO MANY PROGRAMMERS WORK ON PROJECTS THEY NEITHER NEED NOR WANT
@opencontent
RE USE WHEN POSSIBLE
EVERY GOOD PIECE OF WORK STARTS BY SCRATCHING A PERSONAL ITCH
WITH ENOUGH EYEBALLS ALL BUGS ARE SHALLOW
@giuliaforsythe #oped12

CHERI Adoption and Diffusion Research – April 2024

'We have identified several key sectors who would benefit from CHERI adoption including telecoms, automotive, defence, Information Technology (IT), finance, health and utilities. We have prepared case studies for the first four (others faced similar challenges, benefits and enablers for adoption).'

*'The key market segments for CHERI adoption within the semiconductor supply chain are **chip designers** and **systems manufacturers**. Software engineering is also important, as program code needs to be rewritten for the CHERI architecture to ensure the benefits of CHERI are fully realised.'*

Potential future adoption of CHERI

We explored the barriers and enablers to adoption in 5 potential adoption sectors.

Mobile devices: Low likelihood of adoption in the near term

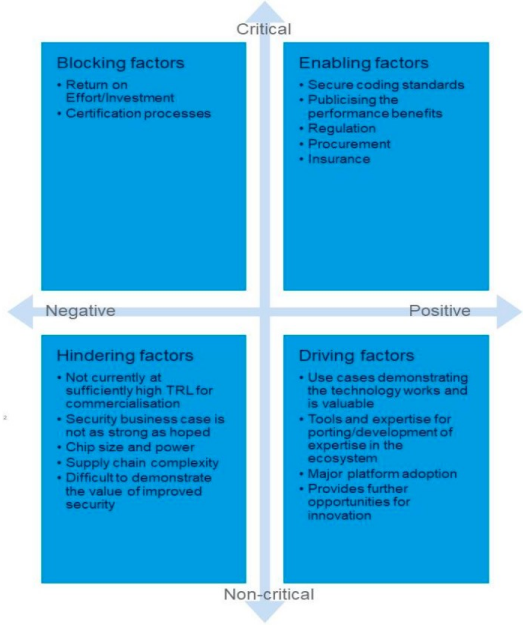
Telecoms Infrastructure: Some strong benefits of adoption, but also some challenges

IOT/Embedded Systems: Economics of adoption for embedded systems are advantageous for CHERI adoption

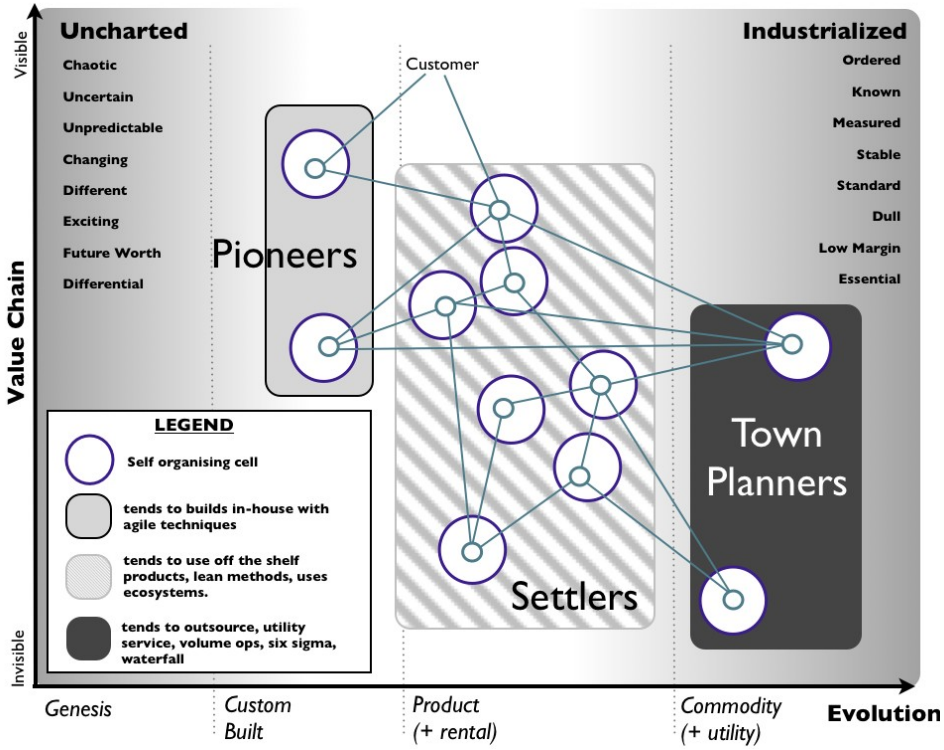
Defence: Evidence of interest from the sector and possible routes to wider adoption through Government procurement levers

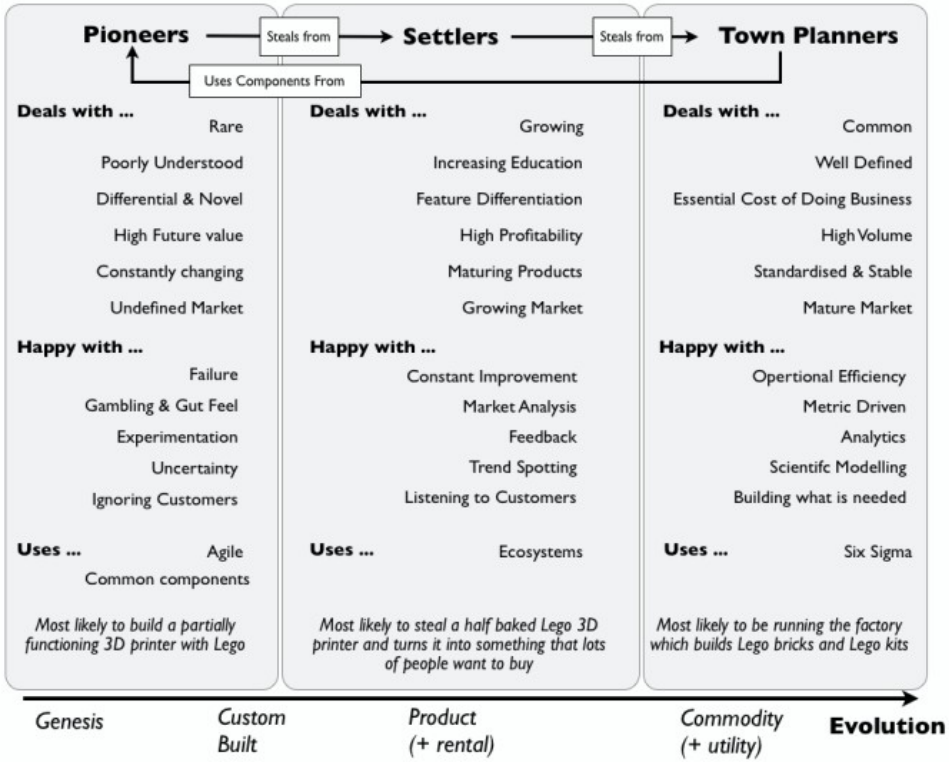
Automotive: Some likelihood of adoption with some challenges to overcome

Figure 11: Summary of Barriers and Enablers to CHERI adoption



How to scale the CHERI innovation ecosystem?





(Source: Simon Wardley, 'Pioneers, Settlers, Town Planners' https://blog.gardeviance.org/2015/03/on-pioneers-settlers-town-planners-and.html)

CyNam connects talented cyber technology minds and businesses so they can Network. Innovate. Collaborate

CYNAM

The Ecosystem

7,000+ Individual community members

15 Cyber-tech businesses from primes to SMEs

tulpa

NORTHROP GRUMMAN

GEMBA ADVANTAGE

CGI

hexiosec

33 Professional services firms supporting the ecosystem

hcrlaw

WYNNE-JONES IP own thinking

hub8

Hazlewoods

INFOSEC

25+ Actively engaged investors

Foresight

mercia Ventures

Amadeus Capital Partners

TENELEVEN

octopus ventures

14 Public sector organisations engaged

Gloucestershire County Council

CHELTENHAM BOROUGH COUNCIL

Department for Business & Trade

National Cyber Security Centre

GCHQ

1 Links with regional UK cyber clusters

CYBER WALES

CYBER. LONDON

UKC3 UK Cyber Cluster Collaboration

SCOTLAND IS

BRISTOL & BATH CYBER

CyberNorth

25+ Academic, education and cultural sector connections

UNIVERSITY OF GLOUCESTERSHIRE

UWE Bristol

CHELTENHAM Festivals

Gloucestershire College

Cyber First

CYNAM

Golden Valley Development

GOLDEN VALLEY

Innovation Centre

This nationally significant asset will extend to approximately 160,000 sq ft and comprise high quality office space (self-contained and co-working suites), flexible event spaces, food & beverage facilities and high-quality public realm.

Over 1000 New Homes

Over 1,000 new homes across a variety of types, tenures, and sizes to meet all needs.

Over 1 Million sq. ft.

Commercial space clustering companies from the tech sectors such as cybersecurity, AI and more – all 100 meters from GCHQ's Doughnut

datacity | Department for Science, Innovation & Technology | cambridge econometrics

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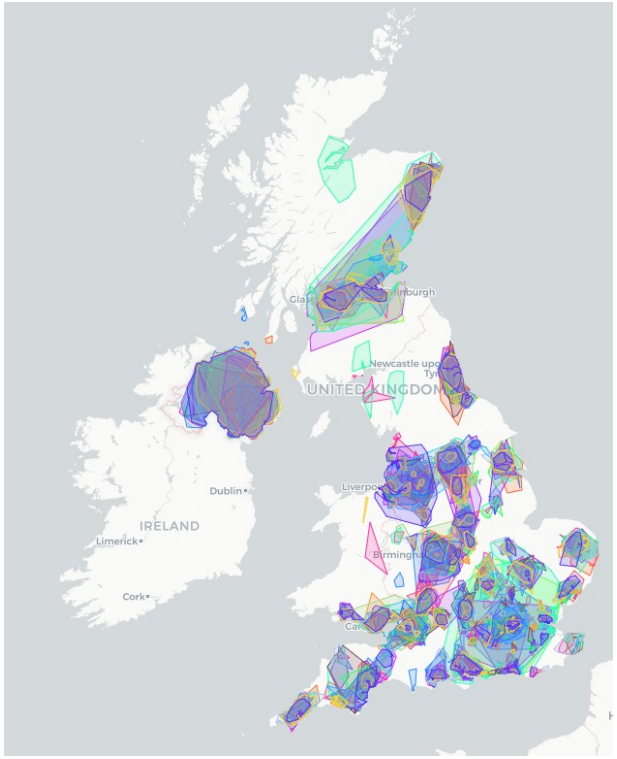
The Innovation Clusters Map from the Department for Science, Innovation and Technology



Bristol & Bath Cyber

Cyber East

CyberNorth

CyNam (Cyber Cheltenham)

Cyber London

Cyber Wales

East Midlands Cyber Security Cluster

Kent & Medway Cyber Cluster

Midlands Cyber (West Midlands)

NI (Northern Ireland) Cyber

North West Cyber Security Cluster

OxCyber

ScotlandIS

Surrey Cyber Security Cluster

South West Cyber Security Cluster

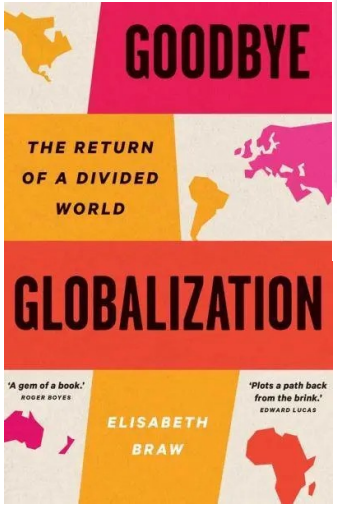
Swindon and Wiltshire Cyber Cluster

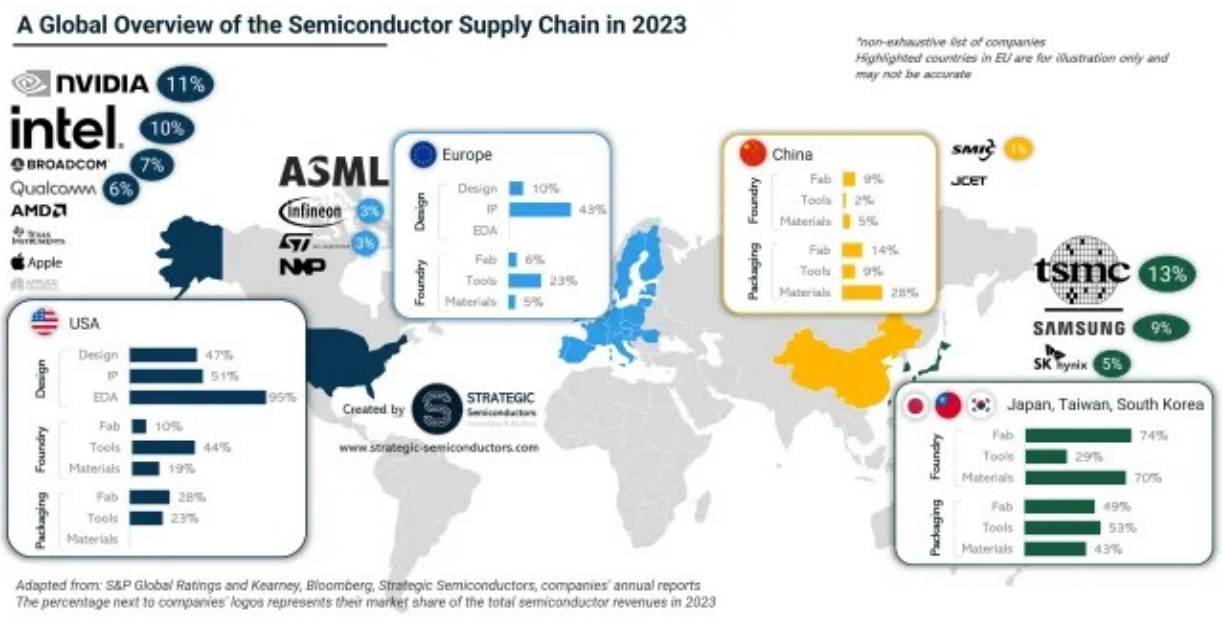
Yorkshire Cyber Security Cluster

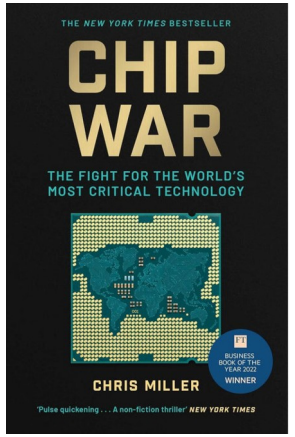
UKC3

UK Cyber Cluster Collaboration

Global growth of CHERI?



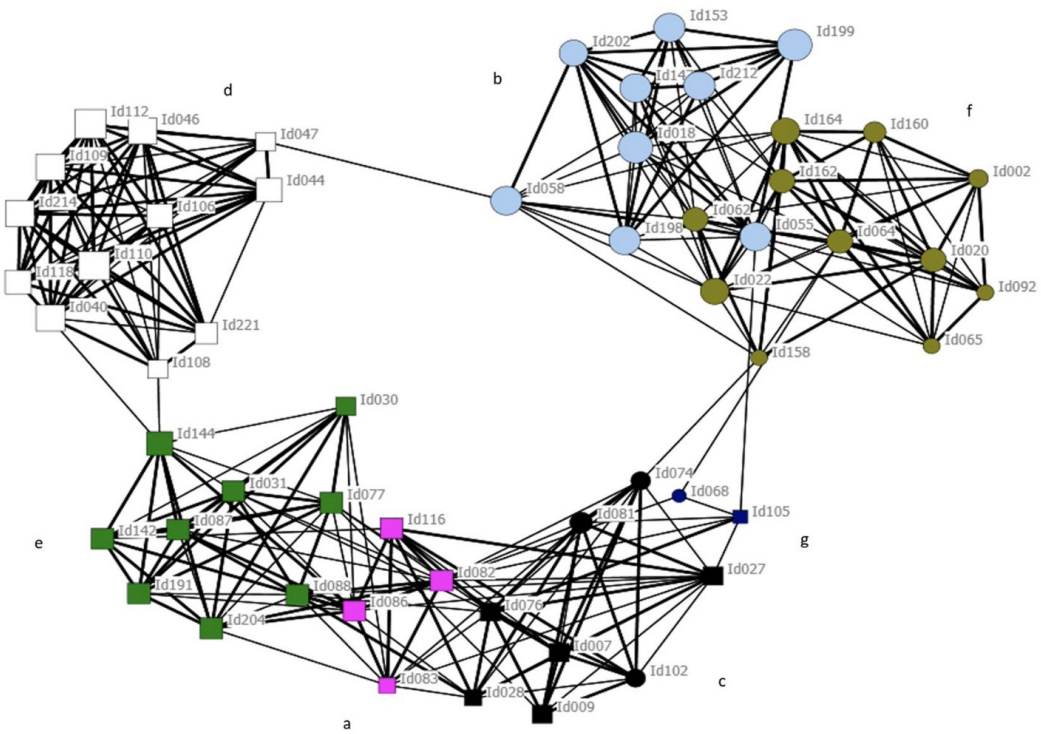


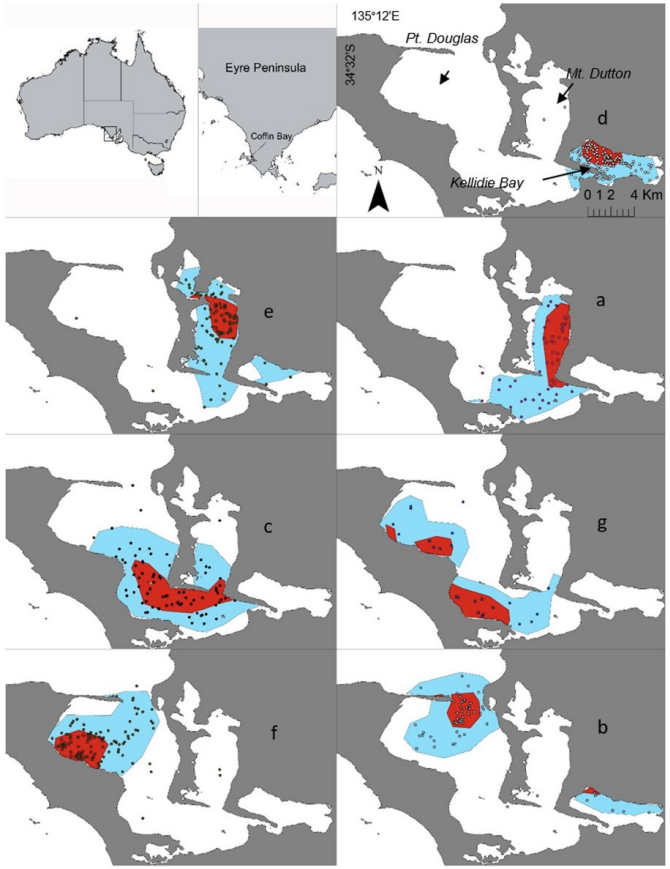


What role will you **play** in building the innovation ecosystem?

Why would dolphins make good ecosystem builders?

Dolphins create social clusters …





Source: 'Kinship and reproductive condition correlate with affiliation patterns in female southern Australian bottlenose dolphins', Diaz-Aguirre et-al, 2020)

THANK YOU!